Multi-Scale Computational Analysis of Alzheimer's Disease Using Transcriptomics and Machine Learning

Integrating Bulk RNA-seq, Single-Cell Transcriptomics, and Predictive Modeling for Molecular Signature Discovery

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Research Area: Computational Biology | Bioinformatics | Alzheimer's Disease Research
Methods: RNA-seq Analysis | Single-Cell Transcriptomics | Machine Learning
Programming & Tools: Python | R | Scanpy | AnnData | pandas | scikit-learn | DESeq2 | rMATS | STAR
Research Experience: University at Albany High School Bioinformatics Program (2026)
Data Sources: Publicly available and program-provided Alzheimer's disease transcriptomic datasets, including bulk RNA-seq and single-nucleus RNA-seq datasets.

Project Overview

This portfolio presents an integrated computational framework investigating Alzheimer's disease (AD) through three complementary projects. By combining bulk RNA sequencing analysis, single-nucleus transcriptomics, and supervised machine learning, this work demonstrates how computational approaches can identify disease-associated molecular signatures, cellular patterns, and predictive features across multiple biological levels.

Abstract

Alzheimer's disease (AD) is a complex neurodegenerative disorder associated with changes in gene expression, cellular function, and molecular pathways. This portfolio investigates AD using an integrated computational approach combining bulk RNA sequencing, single-cell transcriptomics, and machine learning.

Bulk RNA-seq analysis identified differential gene expression, alternative splicing events, including changes in the APP gene, and biological pathways related to neuronal function and cellular communication. Single-cell RNA sequencing revealed distinct brain cell populations and differences in cellular composition across Alzheimer's pathology groups. A logistic regression model trained on single-cell gene expression data achieved 73.1% accuracy in predicting Alzheimer's pathology states and identified predictive genes, including IGLC3, MS4A14, SERPINA1, CCL3, and CCL8. Many of these genes are associated with immune regulation and inflammatory processes.

Together, these analyses demonstrate how computational biology approaches can integrate transcriptomic data, single-cell analysis, and machine learning to identify molecular signatures associated with Alzheimer's disease and provide insight into disease mechanisms.

Research Question

Can integrated computational approaches combining bulk RNA sequencing, single-cell transcriptomics, and machine learning identify molecular signatures associated with Alzheimer's disease and reveal disease-related biological changes?

Project Goals

- Identify gene expression and alternative splicing changes associated with Alzheimer's disease.
- Analyze cellular differences and disease-associated patterns at the single-cell level.
- Apply machine learning approaches to predict Alzheimer's pathology states using gene expression profiles.
- Identify important molecular features that may contribute to disease classification.

Overall Approach

RNA-seq Analysis → Identified gene expression and alternative splicing changes.

Single-Cell Transcriptomics → Examined cellular populations and disease-associated patterns.

Machine Learning → Predicted pathology states and identified predictive genes.

Project 1: RNA-seq and Alternative Splicing Analysis

Objective

To identify gene expression changes and alternative splicing events associated with Alzheimer's disease using bulk RNA sequencing data.

Dataset & Methods

Dataset: Bulk RNA sequencing data from human Alzheimer's disease and control brain samples were analyzed to identify disease-associated changes in gene expression and RNA regulation.

- **Sequence Alignment:** Aligned RNA-seq reads to the human reference genome using **STAR**.
- **Differential Expression Analysis:** Quantified gene expression changes between Alzheimer's disease and control samples using **DESeq2 in R**.
- **Alternative Splicing Analysis:** Identified differential splicing events using **rMATS** and analyzed splicing patterns using **maser**.
- **Pathway Enrichment Analysis:** Performed Gene Ontology analysis using **Metascape** to identify biological pathways affected by Alzheimer's disease.

Key Findings

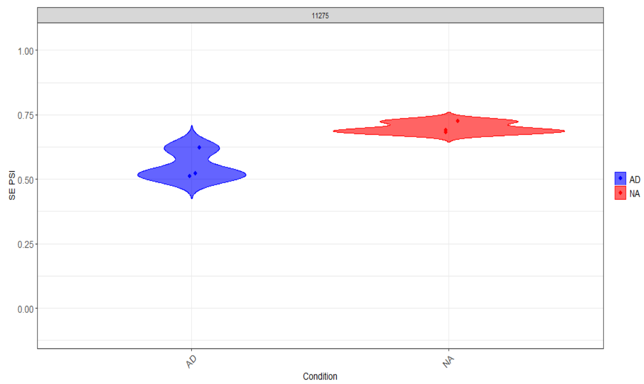
- Identified differentially expressed genes between Alzheimer's disease and control brain samples, revealing transcriptomic differences associated with disease state.
- Detected an alternative splicing event in the APP (Amyloid Precursor Protein) gene, demonstrating that Alzheimer's disease affects RNA processing regulation in addition to changes in overall gene expression.
- Gene Ontology analysis identified altered biological pathways related to neuronal function, synaptic communication, and cellular organization, providing insight into molecular processes affected during Alzheimer's disease.
- Principal component analysis showed separation between Alzheimer's disease and control samples based on global gene expression patterns, supporting distinct transcriptomic profiles between disease and control groups.

Skills & Tools

- **Programming:** R
- **Bioinformatics:** STAR, DESeq2, rMATS, maser
- **Pathway Analysis:** Metascape / Gene Ontology

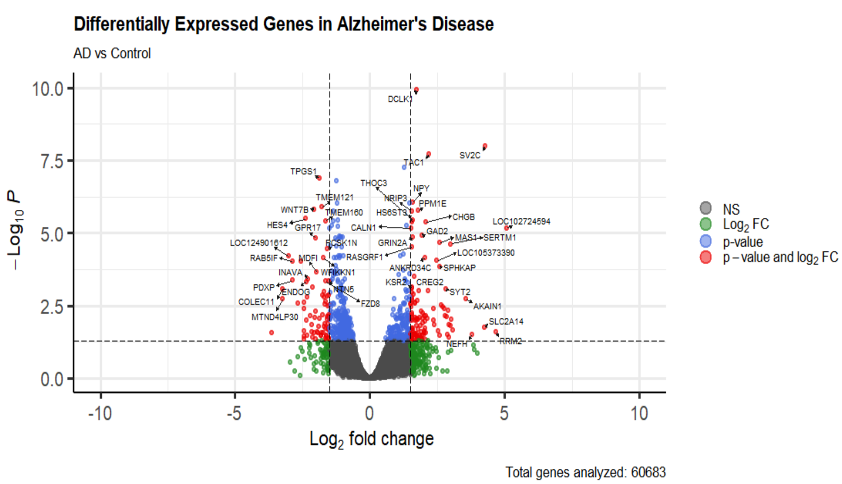
RNA-seq & Alternative Splicing Analysis – Figures

Figure 1. Alternative Splicing Analysis of the APP Gene



Violin plot *displaying the distribution of Percent Spliced In (PSI) values for an exon inclusion event in the Amyloid Precursor Protein (APP) gene. PSI represents the proportion of transcripts containing a specific exon, ranging from 0 to 1 (0% to 100%). Alzheimer's disease (AD) samples show altered PSI values compared to non-AD healthy controls, indicating changes in RNA splicing regulation associated with Alzheimer's disease.*

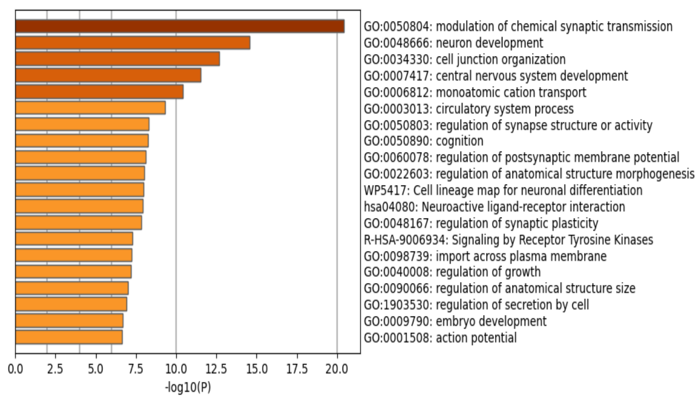
Figure 2. Differential Gene Expression Analysis



Volcano plot *summarizing differential gene expression across 60,683 analyzed genes. The x-axis represents the magnitude of expression change measured by $\log_2$ fold change, while the y-axis represents statistical significance using $-\log_{10}$ p-value. Genes highlighted in red represent statistically significant expression changes with larger differences between Alzheimer's disease and control samples. Labeled genes, including DCLK1 and WNT7B, represent examples of genes showing notable expression changes associated with Alzheimer's disease.*

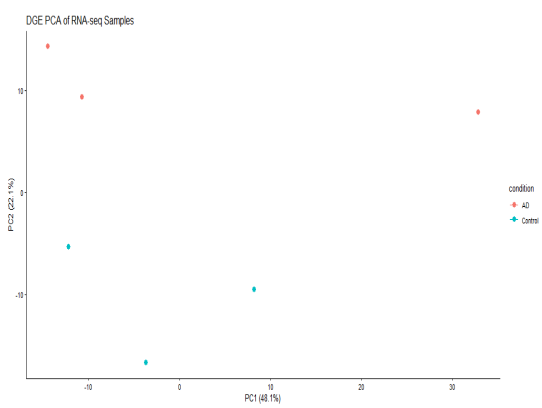
RNA-seq Additional Analysis

Figure 3. Gene Ontology Pathway Analysis



***Bar plot** showing biological pathways enriched among differentially expressed genes identified from Alzheimer's disease and control samples. Gene Ontology analysis groups genes into biological categories to identify cellular processes associated with observed expression changes. The x-axis represents pathway enrichment significance using $-\log_{10}$ p-value, where higher values indicate stronger statistical enrichment. The most significantly enriched pathways included chemical synaptic transmission and neuronal development, highlighting biological processes associated with neuronal communication and brain function in Alzheimer's disease.*

Figure 4. Principal Component Analysis of RNA-seq Samples



***PCA scatter plot** showing separation of Alzheimer's disease (AD) and control samples based on overall gene expression profiles. PCA reduces thousands of gene expression measurements into principal components that capture the largest sources of variation across samples. Principal Component 1 (PC1) explains 48.1% of the total variance, while Principal Component 2 (PC2) explains 22.1%, together representing over 70% of transcriptomic variation. The separation between AD and control samples indicates distinct global gene expression patterns associated with Alzheimer's disease.*

RNA-seq Project Interpretation & Synthesis

Global Transcriptomic Profiles (PCA)

Principal Component Analysis (PCA) showed separation between Alzheimer's disease and control samples based on overall gene expression patterns. PC1 explained **48.1%** of the variance and PC2 explained **22.1%**, together capturing over **70% of transcriptomic variation**. This clustering indicates distinct global gene expression profiles associated with Alzheimer's disease.

Gene Expression & Alternative Splicing Changes

Differential gene expression analysis identified altered expression patterns between Alzheimer's disease and control samples, including notable genes such as **DCLK1** and **WNT7B**. Alternative splicing analysis revealed changes in exon inclusion within the **APP** gene, demonstrating that Alzheimer's disease is associated with changes in RNA processing in addition to gene expression changes.

Biological Pathway Changes

Gene Ontology analysis connected transcriptomic changes to biological processes involved in Alzheimer's disease. Enriched pathways included **chemical synaptic transmission, neuronal development, and synaptic plasticity**, highlighting changes in pathways important for neuronal function.

Transition to Single-Cell Analysis

While bulk RNA sequencing provides a broad view of brain tissue changes, it averages signals across multiple cell types. Single-cell transcriptomics was therefore used in Project 2 to identify which specific cellular populations contribute to Alzheimer's disease-associated molecular patterns.

Project 2: Single-Cell RNA Sequencing Analysis

Objective

To investigate cell-type-specific gene expression patterns and cellular population changes associated with Alzheimer's disease across different pathology stages.

Dataset & Methods

Dataset: Publicly available post-mortem human brain single-nucleus RNA sequencing dataset (**NCBI GEO: GSE243292**) was analyzed to investigate cellular patterns associated with Alzheimer's disease pathology.

- **Quality Control & Processing:** Filtered and normalized single-cell data and selected highly variable genes using **Scanpy in Python**.
- **Dimensionality Reduction:** Performed **PCA** followed by **UMAP** visualization to identify patterns and relationships among individual cells.
- **Cell-Type Annotation:** Identified cell populations using gene expression patterns and known cell-type marker genes.

Key Findings

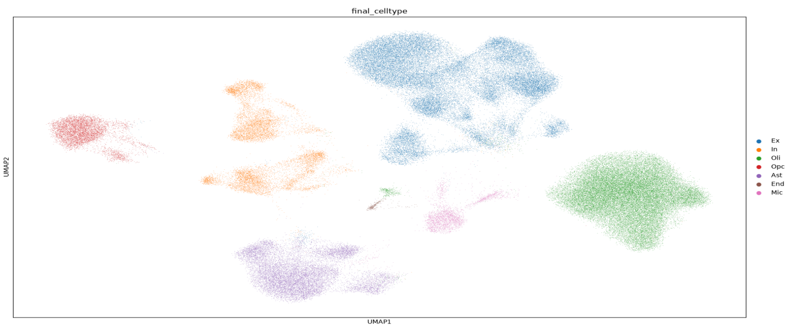
- Identified major brain cell populations, including **excitatory neurons, inhibitory neurons, oligodendrocytes, oligodendrocyte progenitor cells (OPCs), astrocytes, microglia, and endothelial cells**.
- UMAP analysis revealed distinct cellular clusters based on gene expression patterns.
- Cell composition analysis compared populations across Alzheimer's pathology groups, including **A−T− (control), A+T− (amyloid-positive), and A+T+ (amyloid and tau-positive)** samples.
- Observed changes in immune-related cell populations, including **microglia and astrocytes**, highlighting their potential involvement in Alzheimer's disease-associated changes.

Skills & Tools

- **Programming:** Python
- **Single-Cell Analysis:** Scanpy, AnnData, Pandas
- **Methods:** Quality control, normalization, highly variable gene selection, PCA, UMAP, clustering, cell-type annotation, cellular composition analysis

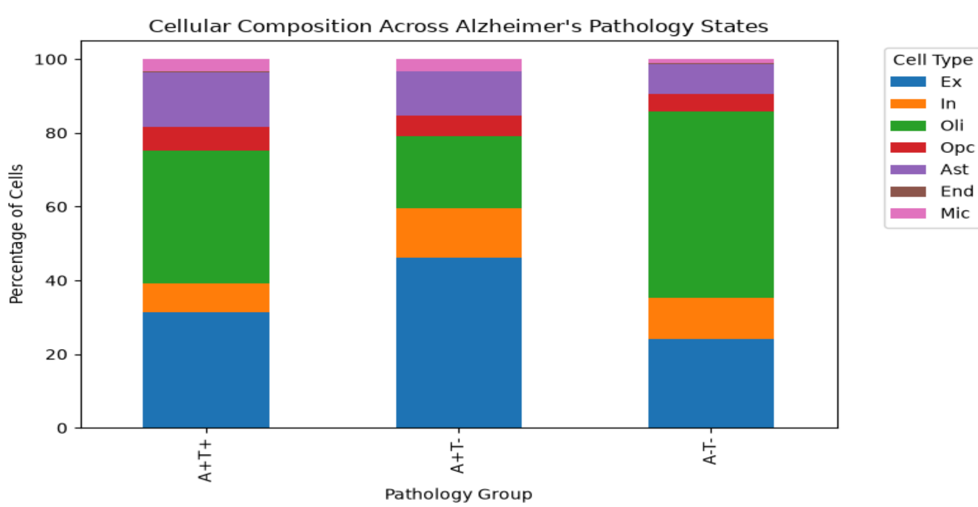
Single-Cell RNA Sequencing Analysis – Figures

Figure 5. UMAP Visualization of Single-Cell Transcriptomic Profiles



Two-dimensional UMAP *visualization showing individual cell nuclei grouped based on transcriptomic similarity. UMAP reduces thousands of gene expression measurements into a two-dimensional representation where cells located closer together have more similar expression patterns. Cells were annotated into major brain cell populations, including Excitatory Neurons (Ex), Inhibitory Neurons (In), Oligodendrocytes (Oli), Oligodendrocyte Progenitor Cells (Opc), Astrocytes (Ast), Endothelial Cells (End), and Microglia (Mic).*

Figure 6. Cellular Composition Across Alzheimer's Pathology Groups



Stacked bar graph *showing the relative abundance of major brain cell populations across Alzheimer's pathology groups:* ***A−T− (no detected amyloid or tau pathology), A+T− (amyloid-positive and tau-negative), and A+T+ (amyloid-positive and tau-positive)*** *samples. Oligodendrocytes (Oli) and Excitatory Neurons (Ex) represent major cell populations across groups, while differences in glial populations, including Astrocytes and Microglia, highlight changes in cellular composition associated with Alzheimer's pathology.*

Single-Cell RNA-seq Project Interpretation & Synthesis

Cellular Heterogeneity and Clustering (Figure 5)

Single-cell transcriptomic analysis revealed the cellular complexity of Alzheimer's disease brain tissue by identifying seven major cell populations: **Excitatory Neurons, Inhibitory Neurons, Oligodendrocytes, OPCs, Astrocytes, Microglia, and Endothelial cells**. UMAP visualization (Figure 5) showed distinct transcriptional clusters based on gene expression patterns, allowing individual cell populations to be examined separately.

Cellular Composition Across Alzheimer's Pathology Groups (Figure 6)

Cell composition analysis (Figure 6) compared cellular populations across different Alzheimer's pathology groups, including **A−T−, A+T−, and A+T+** samples. Differences in the proportions of glial populations, including **microglia and astrocytes**, highlight changes in immune-related and supportive cell populations associated with Alzheimer's disease.

Bulk RNA-seq vs. Single-Cell Resolution

While bulk RNA sequencing provides a broad view of tissue-level gene expression and RNA splicing changes, it averages signals across multiple cell types. Single-cell transcriptomics improves resolution by identifying which specific cellular populations contribute to observed molecular patterns.

Transition to Machine Learning Classification

The identification of distinct single-cell expression patterns raises an important computational question: Can these gene expression profiles be used to predict Alzheimer's pathology states? In Project 3, machine learning approaches were applied to classify pathology groups and identify predictive molecular features.

Project 3: Machine Learning Prediction of Alzheimer’s Pathology

Objective

To determine whether machine learning models can predict Alzheimer’s disease pathology states using single-cell gene expression patterns.

Dataset & Methods

Single-cell RNA sequencing data were used as input for machine learning analysis to classify Alzheimer’s pathology states.

- Quality-controlled data were processed by selecting **2,000 highly variable genes** as model features.
- A **Logistic Regression classifier** was trained using an 80/20 train-test split.
- Model performance was evaluated using test accuracy and confusion matrix analysis.
- Feature importance analysis was used to identify genes contributing most to classification.

Key Findings

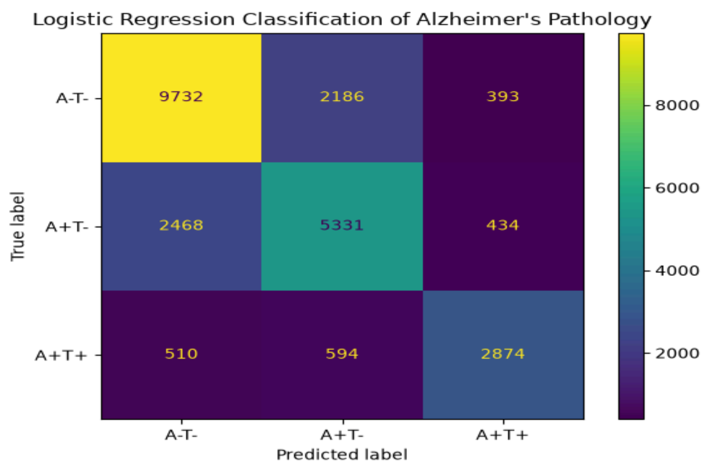
- The Logistic Regression model achieved **73.1% accuracy** in predicting Alzheimer’s pathology states using single-cell gene expression profiles.
- Feature importance analysis identified predictive genes including **IGLC3, MS4A14, SERPINA1, CCL3, and CCL8**, which contributed to distinguishing between pathology groups.
- Several identified genes are associated with immune-related processes. **CCL3 and CCL8** are chemokines involved in immune cell signaling, while **MS4A14 and SERPINA1** have been linked to inflammatory and immune-related responses. These findings suggest that immune-associated molecular patterns contribute to differences observed between Alzheimer’s pathology states.
- The results demonstrate how machine learning can identify biologically meaningful patterns within complex single-cell transcriptomic datasets.

Skills & Tools

- **Programming:** Python
- **Machine Learning:** scikit-learn, Logistic Regression
- **Single-Cell Analysis:** Scanpy, AnnData
- **Data Processing:** Pandas
- **Methods:** Feature selection, model training, classification, feature importance analysis, data processing

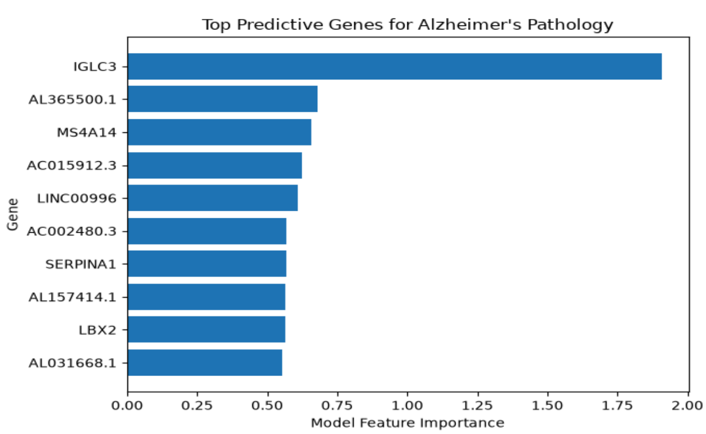
Machine Learning Results – Figures

Figure 7. Machine Learning Model Performance: Confusion Matrix



Confusion matrix *evaluating Logistic Regression classification performance across three Alzheimer's pathology groups: A−T−, A+T−, and A+T+. Rows represent the true pathology labels and columns represent the predicted labels. Values along the diagonal represent correct classifications, while off-diagonal values represent misclassified samples. The model achieved an overall accuracy of 73.1%, demonstrating that single-cell gene expression patterns contain information that can distinguish between Alzheimer's pathology states.*

Figure 8. Top Predictive Genes Identified by Machine Learning



Feature importance ranking *of the top genes contributing to Logistic Regression classification based on the magnitude of model coefficients. Genes with larger coefficients had a greater influence on predicting Alzheimer's pathology states. The top predictive genes included IGLC3, AL365500.1, MS4A14, AC015912.3, and LINC00996. Among these, IGLC3 is associated with immune signaling, while MS4A14 belongs to the MS4A gene family involved in immune and microglial-related processes. These findings highlight immune-associated gene expression patterns as important features for distinguishing Alzheimer's pathology states.*

Interpretation of Machine Learning Results

Predictive Power of Single-Cell Profiles

Machine learning analysis demonstrated that single-cell gene expression patterns contain information capable of distinguishing between Alzheimer's pathology groups (**A−T−, A+T−, and A+T+**). A Logistic Regression model achieved **73.1% accuracy**, showing that selected gene expression features captured meaningful molecular differences associated with Alzheimer's pathology.

Biological Meaning of Predictive Genes

Feature importance analysis identified genes contributing to classification, including **IGLC3, MS4A14, SERPINA1, CCL3, and CCL8**. Several of these genes are associated with immune-related processes: **CCL3 and CCL8** function as chemokines involved in immune signaling, while **MS4A14 and SERPINA1** have been linked to inflammatory and immune responses. These findings suggest that immune-associated gene expression patterns contribute to differences observed across Alzheimer's pathology groups.

Integrated Multi-Scale Analysis

Together, the three projects provide a multi-scale computational view of Alzheimer's disease:

1. Bulk RNA-seq: Identified tissue-level gene expression changes, alternative splicing alterations in **APP**, and disrupted biological pathways related to neuronal function.

2. Single-cell RNA-seq: Resolved these molecular changes at the cellular level by identifying specific brain cell populations and differences in cellular composition across pathology groups.

3. Machine Learning: Used single-cell gene expression patterns to classify pathology states and identify predictive molecular features associated with disease-related changes.

Overall, this portfolio demonstrates how integrating transcriptomics, single-cell analysis, and machine learning can uncover molecular patterns associated with Alzheimer's disease and provide a framework for discovering potential biomarkers for future investigation.

Research Skills & Computational Tools

Programming & Data Analysis

Python

- Single-nucleus RNA sequencing analysis using **Scanpy and AnnData**
- Data preprocessing, quality control, normalization, and visualization
- Machine learning workflows using **scikit-learn**

R

- Differential gene expression analysis using **DESeq2**
- Statistical analysis and visualization
- Alternative splicing analysis using **maser**

Bioinformatics Workflows

Bulk RNA-seq Analysis

- RNA-seq read alignment using **STAR**
- Differential expression analysis with **DESeq2**
- Alternative splicing identification using **rMATS and maser**
- Gene Ontology pathway enrichment using **Metascape**

Single-Cell Transcriptomics

- Quality control and normalization of single-cell datasets
- Highly variable gene selection
- PCA and UMAP dimensional reduction
- Cell-type identification using gene expression markers

Machine Learning & Predictive Modeling

- Feature selection using highly variable genes
- Logistic Regression classification of Alzheimer’s pathology states
- Model evaluation using accuracy and confusion matrices
- Identification of predictive molecular features through feature importance analysis

Scientific Research Skills

- Analysis of large-scale genomic datasets containing **over 60,000 genes**
- Interpretation of computational results in the context of Alzheimer’s disease biology
- Scientific communication through research posters and written reports

Future Directions / Continued Research

The integrated analyses presented in this portfolio demonstrate how bulk RNA sequencing, single-cell transcriptomics, and machine learning can be combined to investigate molecular patterns associated with Alzheimer's disease. Future work could expand these findings through additional computational and biological approaches:

- **Advanced Machine Learning Models:** Test additional classification approaches, including ensemble methods and neural networks, to determine whether more complex models improve prediction of Alzheimer's pathology states.
- **Independent Dataset Validation:** Validate predictive genes, including **IGLC3, MS4A14, SERPINA1, CCL3, and CCL8**, using additional Alzheimer's disease datasets to evaluate whether these molecular patterns are consistent across cohorts.
- **Cellular Interaction Analysis:** Explore cell-cell communication and regulatory networks to better understand interactions between microglia, astrocytes, and neuronal populations.
- **Brain Region and Spatial Analysis:** Analyze additional brain regions, such as the **hippocampus and entorhinal cortex**, and incorporate spatial transcriptomics approaches to study how molecular changes are organized within brain tissue.
- **Expanding Computational Biology Skills:** Continue developing skills in bioinformatics, high-throughput genomic analysis, and artificial intelligence applications in biomedical research.

Acknowledgements

- I would like to thank the **University at Albany Bioinformatics Program** for providing the opportunity to conduct computational biology research and gain hands-on experience with transcriptomic analysis, single-cell sequencing, and machine learning approaches.
- I am grateful for the guidance and support from the instructors and mentors who helped develop my skills in bioinformatics, data analysis, and scientific communication throughout this research experience.
- I also appreciate the publicly available genomic datasets and open-source computational tools that made this analysis possible and allowed me to explore the application of computational methods in Alzheimer's disease research.

Reproducibility

To ensure reproducibility, computational analyses and project files were organized and archived. The workflow includes R-based scripts and workspaces for bulk RNA-seq differential expression and alternative splicing analyses, Python scripts for single-cell RNA sequencing and machine learning, processed result files, selected generated figures, and command histories.

The complete analysis repository containing computational workflows, scripts, and supporting files is available on GitHub:

GitHub Repository: https://github.com/siddharthvivek23/Alzheimer-Transcriptomic-Analysis

Public datasets used in this project include program-provided and publicly available Alzheimer's disease transcriptomic datasets, including the single-cell dataset GSE243292 from the NCBI Gene Expression Omnibus (GEO). These resources provide the materials necessary to reproduce and extend the computational analyses presented in this portfolio.

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